Nathan Kho Pancras

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Summary

Final-year Informatics Technology student (18) with keen interest and active study in **cybersecurity**, including **cryptography**, **forensics**, and **red teaming** while being knowledgeable in other areas. Skilled in C++ and Python, and experienced in full-stack development with extra knowledge in robotics development and AI.

Education

Sepuluh Nopember Institute of Technology *2023 - 2027 (Expected)*
B.Sc. in Information Technology

- CGPA: 3.90/4.0
- Assistant Lecturer for Information Asset Protection, Operating Systems, Data Structures and OOP, Ethical Hacking and Penetration Testing, Cloud Computing Technology (Spring 2025 – Spring 2026)

Professional Highlights

Security Engineer Intern *Remote*
Xendit *May 2026 - Present*

- Executed internal penetration tests across corporate infrastructure to proactively identify, assess, and remediate security vulnerabilities.
- Developed and implemented multiple automated security workflows to streamline internal threat assessments and operational processes.
- Administered the company's private bug bounty program, triaging vulnerability submissions from external researchers.

Summer Research Intern *Singapore*
National University of Singapore *June 2026 - Aug 2026*

- Co-authored research on fingerprinting autonomous AI agents from encrypted network traffic by analyzing TLS packet metadata without payload decryption.
- Designed a Docker-based cyber range testbed to securely capture and evaluate agent behavior under diverse network conditions.
- Engineered an ML classification pipeline using 61 statistical features, achieving an 85% accuracy rate in attributing LLM backends based on inference timing.
- Conceptualized and developed practical cybersecurity challenges for the university's CS4238 course to support academic instruction.

Cybersecurity Competitions

- Cryptography stream lead and player in the cryptography / forensics stream for HCS since 2023, ITS's cybersecurity club, currently ranked as Indonesia's fifth best CTF team.

Cybersecurity Challenge Developer

Designed and implemented intricate cybersecurity challenges (CTF) for major events, providing real-time support to participants.

- A Renewal Agent (ARA) CTF 6.0 - Indonesia - 2 challenges in cryptography
- Badge to Breach: ICS Cyber Siege 2025 - Malaysia - 1 challenge in cryptography
- Schematics NPC CTF 2025 - Indonesia - 3 challenges in cryptography
- A Renewal Agent (ARA) 7.0 CTF 2026 - Indonesia - 2 challenges in cryptography

Software Engineer / Programmer *Surabaya*
Banyubramanta ITS *2024 - 2026*

- Improved a system for an ROV/AUV using STM32 with ROS2, OpenCV, and YOLOv5, created a UI for rapid construction of ROV/AUV behavior trees.

Projects

Portfolio Website - Astro, TailwindCSS, MDX *etern1ty.site*

- Created a personal portfolio and blog page using Astro, TailwindCSS, and MDX.

ECC Digital Signing System - Python, React w/ TailwindCSS *GitHub*

- Designed and implemented a secure digital signing system using **ECDSA-SECP256k1** for robust binary file authentication, featuring AES-256-GCM encrypted key storage.

Q-Sealnet (PQC Signer) - Python, Next.js w/ TailwindCSS *GitHub*

- Designed and implemented a post-quantum digital signing system utilizing **ML-DSA variants (44, 65, 87)** for robust binary file authentication, featuring AES-256-GCM encrypted private-key storage.

Skills and Technologies

Languages: Python, C, C++, SQL, Javascript, Typescript, Solidity

Frameworks: Flask, Express, React, Next.js, Vue.js, Astro, Foundry

Awards and Achievements

- **2nd Place - ARKAVIDIA 10.0 CTF** Mar 2026
  Awarded by: Institut Teknologi Bandung (National)
- **3rd Place - HOLOGY 8.0 & 9.0 CTF** Oct 2025 & Sep 2026
  Awarded by: Universitas Brawijaya (National)
- **2nd Place - HackToday CTF 2025** Sep 2025
  Awarded by: Institut Pertanian Bandung (National)
- **3rd Place - Meta4Sec Cyber Ops Clash CTF 2.0** Aug 2025
  Awarded by: Meta4Sec & redlimit. (National)
- **Top 5 - Standoff 15 for Blue Teams** Apr 2025
  Awarded by: Standoff 365 Team & Positive Technologies (Global)
- **Best Writeup & Top 10 - RECURSION CTF 1.0** Apr 2025
  Awarded by: Universitas Hasanuddin (National)
- **2nd Place - AI vs Human CTF** Mar 2025
  Awarded by: HackTheBox (Global)
- **3rd Place - National Cyber Week CTF** Nov 2024
  Awarded by: BINUS University (National)
- **1st Place / Best Strategy - Kontes Robot Bawah Air Indonesia (KRBAI)** Jun 2024
  Awarded by: Puspresnas / Ministry of Education, Culture, Research, and Technology (National)

Other Awards

- Finalist - GEMASTIK Division II (Cyber Security) 2024 & 2025
- Top 10 - National Cyber Week CTF 2025
- Top 10 - Cyber Breaker Development (CBD) CTF 2025
- Top 10 - ARKAVIDIA 9.0 CTF 2025
- Top 4 - Netcomp 3.0 CTF 2024

Committees

- **Head of IT Olympiad** (ARA 7.0) Sep 2025 – Mar 2026
- **Expert Staff of IT Olympiad** (ARA 6.0) Oct 2024 – May 2025
- **Staff of IT Olympiad** (ARA 5.0) Oct 2023 – May 2024